

Homework 8

1. Consider in-plane displacements for the M_x point (i.e. $\mathbf{k} = (\frac{1}{2}, \frac{1}{2})$) in graphene. This problem parallels what we did in lecture for the K-point.
 - (a) Derive the composition of the atom representation.
 - (b) Derive the composition of the polarization representation.
 - (c) Derive the composition of the full representation.
 - (d) Use the projection operator to construct all irrep vectors of the full representation.
 - (e) Create a sketch of the identity irrep vector in the M_x supercell.
 - (f) Using the γ_σ and γ_π springs from last problem set, derive an analytic equation for the frequencies of each irrep vector. This should reproduce your numerical result from the last problem set.